Naming Compounds - (Chemical Formulae)

Prefixes

Mono _____	Di _____	Tri _____
Tetra _____	Pent _____	Hex _____
7: _____	8: _____	9: _____
10: _____		

Naming covalent compounds of two non-metal elements

1. Covalent compounds are combinations of two or more non-metal elements. They are not ionic.
2. These compounds consist of molecules (eg, H_2O, CO_2, CO, SO_3) in which atoms are sharing electrons so that each one has a full valence shell.
3. Some covalent compounds have "common names" that do not follow any particular set of rules. These must be learned - for example:

Compound	Common name	Compound	Common name
H_2O	water	$CHCl_3$	chloroform
CH_4	methane	HCl	hydrogen chloride
NH_3	ammonia	HBr	hydrogen bromide
H_2O_2	hydrogen peroxide	HF	hydrogen fluoride

4. For most other covalent compounds there are a number of rules to follow for naming them.
5. **Naming rules (nomenclature)**
   1) The name of the first element in the formula is the same as the element itself. (eg; CO_2 is **<u>carbon</u>** dioxide)
   2) The name of the second element in the formula has its ending changed to IDE (eg; CO_2 is carbon di**<u>oxide</u>**)
   3) If there are <u>more than one</u> of the <u>first element's</u> atoms, the prefixes di (2), tri (3), tetra (4), penta (5), hexa (6), hepta (7), octa (8), nona (9), deca (10) are used to show how many atoms.

Dioxide is easier to say than decaoxide. Try saying both!

there are.

(eg; P_4O_{10} is **tetra**phosphorus decoxide)

4) For the second element, the prefixes mono (1), di (2), tri (3), tetra (4), penta (5), hexa (6), hepta (7), octa (8), nona (9), deca (10) are used to show how many atoms there are.

(eg; P_4O_{10} is tetraphosphorus **dec**oxide)

Name these covalent compounds	
CO	
SO_2	
CO_2	
SO_3	
NO_2	
N_2O_3	
N_2O	
N_2O_5	
Cl_2O_7	
Br_2O_5	
CH_4	
P_2O_3	
P_2O_5	
P_4O_6	
NH_3	
SiF_4	
BF_3	
SF_6	
BrI_5	
OBr_2	
$SiCl_4$	
OF_2	
HCl	
H_2O_2	
H_2O	

$SeCl_2$	
NF_3	
PCl_5	
SiI_4	
IF_7	
B_4O_6	

Write formulas for these covalent compounds

Hydrogen sulfide	
Dinitrogen tetrachloride	
Sulfur pentabromide	
Selenium hexafluoride	
Nitrogen triiodide	
Tetraphosphorus hexasulfide	
Tetraphosphorus decasulfide	
Chlorine difluoride	
Oxygen dibromide	
Chlorine pentabromide	
Silicon tetrabromide	
Silicon tetrafluoride	
Disilicon hexachloride	
Disilicon hexabromide	
Methane	
Ammomia	
Sulfur tetrachloride	
Dinitrogen trisulfide	
Dibromine monoiodide	
Diselenium pentabromide	
Trisilicon octafluoride	
Tetrasilicon decachloride	

Diphosphorus tetrabromide	
Tellurium hexafluoride	
Tellurium trioxide	
Tellurium hexabromide	
Hexaphosphorus nonoxide	
Dicarbon trioxide	
Trisilicon diphosphide	
Boron monophosphide	
Astatine pentafluoride	

Naming Ionic compounds of one metal and one non-metal

1. Ionic compounds are formed between metals (left on the periodic table) and non-metals (right on the periodic table).
2. These compounds consist of a lattice of positive and negative ions. (eg, NaCl, $AgNO_3$, Fe_2O_3) is which the metal has "donated" one or more eletrons to become positive, whilst the non-metal has accepted electrons to become negative.
3. Some ionic compounds have "common names" that do not follow any particular set of rules. These must be learned - for example:

Compound	**Common name**	**Compound**	**Common name**
Fe_2O_3	rust		
NaCl	Salt or table salt		

4. For most other covalent compounds there are a number of rules to follow for naming them.
5. **Naming rules (nomenclature)**

i) The Metal is named first and does not have its name changed, so it is written as seen in the periodic table.

ii) The charge on the metal is indicated by roman numerals in brackets after its' name. This is not necessary for metals that only exist in one state. (eg. Sodium is written as Na, because it only comes as Na^+, whilst iron would have to be written as Fe(II) or Fe(III), as it exists as Fe^{2+} and Fe^{3+}.

iii) The non-metal ion has a suffix added which indicates its' charge or form. Suffixes may be –ide, -ite, -ate, depending on the number of oxygen atoms that are included with the element.

per-	-ate greatest number of oxygen atoms
	-ate greater
	-ite smaller
hypo-	-ite smallest number of oxygen atoms

iv) At this stage you will be using the list of ions, periodic tables and all notes in order to determine the names and formula/symbols for the ion.

Name the following ionic substances.

Formula	Name
$NaBr$	
K_2CO_3	
$Mg_3(PO_4)_2$	
$ZnSO_4$	
$Fe(NO_3)_2$	
$Fe(NO_3)_3$	
$AlCl_3$	
$Cr(OH)_3$	
Cu_2S	
$NiSO_3$	
NH_4HCO_3	
Na_2O	
$Sn(CH_3COO)_2$	
SnI_4	
$AgNO_2$	
$LiHSO_4$	
Au_3PO_4	
$CuCN$	
$Cu(CN)_2$	
$Sr(MnO_4)_2$	
MnO	
$MnCl_2$	
$MnCl_4$	
$NaOH$	
$HgSO_4$	
NaH	
CaH_2	
AlH_3	
CoS	
Zn_3N_2	

Write formulas for the following ionic substances

Name	Formula
Sodium carbonate	
Calcium hydrogencarbonate	
Aluminium oxide	
Iron (II) sulfate	
Iron (III) sulfide	
Tin (II) hydroxide	
Tin (IV) nitrate	
Mercury (II) cyanide	
Aluminium oxalate	
Aluminium nitride	
Lead chromate	
Zinc dichromate	
Gold (III) ethanoate (acetate)	
Potassium hydride	
Lithium oxide	
Silver bromide	
Copper (I) sulfite	
Copper (II) sulfate	
Nickel (II) iodide	
Cobalt (II) fluoride	
Nickel (IV) oxide	
Cobalt (III) oxide	
Sodium phosphate	
Ammonium hydrogensulfate	
Calcium oxide	
Manganese (II) oxide	
Manganese (IV) oxide	
Iron (III) phosphate	
Potassium nitride	
Tin (IV) nitride	